Partial Differential Equations: My Notes

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Chapter 1

Basic Boundary Value Problems and ODEs

1.1 Introduction

Let's review the basics of hyperbolic functions, a useful tool to help with solving ODEs. By definition,

$$\begin{aligned}
\cosh(x) &= \frac{e^x + e^{-x}}{2} \\
\sinh(x) &= \frac{e^x - e^{-x}}{2}
\end{aligned}$$

Suppose we had the following second-order ODE,

$$\frac{d^2y}{dt^2} - \alpha^2 y = 0$$

We know from basic ODE theory that,

$$\boxed{y = c_1 e^{\alpha t} + c_2 e^{-\alpha t}}$$

We add 0 to the equation in hopes of rewriting this in terms of the hyperbolic functions.

$$\begin{aligned}
y &= c_1 \left[\frac{e^{\alpha t}}{2} + \frac{e^{\alpha t}}{2}\right] + c_2 \left[\frac{e^{-\alpha t}}{2} + \frac{e^{-\alpha t}}{2}\right] + \underbrace{c_1 \left[\frac{e^{-\alpha t}}{2} - \frac{e^{-\alpha t}}{2}\right]}_{0} + \underbrace{c_2 \left[\frac{e^{\alpha t}}{2} - \frac{e^{\alpha t}}{2}\right]}_{0} \\
&= c_1 \left[\frac{e^{\alpha t}}{2} - \frac{e^{-\alpha t}}{2}\right] + c_1 \left[\frac{e^{\alpha t}}{2} + \frac{e^{-\alpha t}}{2}\right] - c_2 \left[\frac{e^{\alpha t}}{2} - \frac{e^{-\alpha t}}{2}\right] + c_2 \left[\frac{e^{\alpha t}}{2} + \frac{e^{-\alpha t}}{2}\right] \\
&= c_1 \sinh(\alpha t) + c_1 \cosh(\alpha t) - c_2 \sinh(\alpha t) + c_2 \cosh(\alpha t) \\
&= \overline{c_1} \sinh(\alpha t) + \overline{c_2} \cosh(\alpha t)
\end{aligned}$$

$$\boxed{y = \overline{c_1}\sinh(\alpha t) + \overline{c_2}\cosh(\alpha t)}$$

1.2 Some boundary value problems

Example 1

Consider the following problem,

$$\frac{d^2y}{dt^2} = -\lambda y, \quad \text{BCs: } y(0) = 0,\ y(2\pi) = 0$$

The characteristic polynomial for this differential equation is $r^2 + \lambda = 0$, with roots $r = \pm\sqrt{-\lambda}$.

There are three cases that we must consider: $\lambda > 0$, $\lambda = 0$, $\lambda < 0$. We will soon see that $\lambda > 0$ is the only one that produces non-trivial solutions.

We consider three cases for the parameter λ.

1. **Case $\lambda > 0$.**:

$$\begin{aligned} r &= \pm i\sqrt{\lambda} \\ y &= c_1 \cos\left(t\sqrt{\lambda}\right) + c_2 \sin\left(t\sqrt{\lambda}\right) \end{aligned}$$

From the boundary conditions,

$$\begin{aligned} y(0) &= c_1 = 0 \\ y(2\pi) &= c_2 \sin\left(2\pi\sqrt{\lambda}\right) = 0 \end{aligned}$$

Recall that sine function equals 0 when the argument is πn,

$$\begin{aligned} 2\pi\sqrt{\lambda} &= \pi n \\ \sqrt{\lambda} &= \frac{n}{2} \end{aligned}$$

$$\boxed{y_n(t) = \sin\left(t\sqrt{\lambda_n}\right)}$$

2. **Case $\lambda = 0$.**

$$\begin{aligned} y'' &= 0 \\ y' &= c_1 \\ y &= c_1 t + c_2 \end{aligned}$$

From the boundary conditions,

$$\begin{aligned} y(0) &= c_2 = 0 \\ y(2\pi) &= 2\pi c_1 + c_2 = 0 \longrightarrow c_1 = 0 \end{aligned}$$

The only solution is the trivial solution, $y = 0$.

3. **Case $\lambda < 0$.**

$$\begin{aligned} r &= \pm\sqrt{-\lambda} \\ y &= c_1 \cosh\left(t\sqrt{-\lambda}\right) + c_2 \sinh\left(t\sqrt{-\lambda}\right) \end{aligned}$$

From boundary conditions,

$$\begin{aligned} y(0) &= c_1 = 0 \\ y(2\pi) &= c_2 \sinh\left(2\pi\sqrt{-\lambda}\right) = 0 \longrightarrow c_2 = 0 \end{aligned}$$

This also yields the trivial solution.

The solution to the BVP is,

$$\boxed{y = \sum_{n=1} a_n \sin\left(\frac{nt}{2}\right)}$$

Example 2

Consider the following problem,

$$\frac{d^2y}{dt^2} + \lambda y = 0, \quad \text{BCs: } y'(0) = 0,\ y'(2\pi) = 0$$

Let's do a similar process as Example 1.

1. **Case $\lambda > 0$.**:

$$\begin{aligned}
r &= \pm i\sqrt{\lambda} \\
y &= c_1 \cos\left(t\sqrt{\lambda}\right) + c_2 \sin\left(t\sqrt{\lambda}\right) \\
y' &= -c_1\sqrt{\lambda}\sin\left(t\sqrt{\lambda}\right) + c_2\sqrt{\lambda}\cos\left(t\sqrt{\lambda}\right) \\
y'(0) &= c_2\sqrt{\lambda} = 0 \longrightarrow c_2 = 0 \\
y'(2\pi) &= -c_1\sqrt{\lambda}\sin\left(2\pi\sqrt{\lambda}\right) = 0 \\
2\pi\sqrt{\lambda} &= \pi n \\
\sqrt{\lambda} &= \frac{n}{2}
\end{aligned}$$

$$\boxed{y_n = \cos\left(\frac{nt}{2}\right)}$$

2. **Case $\lambda = 0$.**:

$$\begin{aligned}
y'' &= 0 \\
y' &= c_1 \\
y &= c_1 t + c_2 \\
y'(0) &= 0 \longrightarrow c_1 = 0
\end{aligned}$$

$$\boxed{y_n = c_n = \text{constant}}$$

3. **Case $\lambda < 0$.**:

$$\begin{aligned}
r &= \pm\sqrt{-\lambda} \\
y &= c_1 \cosh\left(t\sqrt{-\lambda}\right) + c_2 \sinh\left(t\sqrt{-\lambda}\right) \\
y' &= c_1\sqrt{-\lambda}\sinh\left(t\sqrt{-\lambda}\right) + c_2\sqrt{-\lambda}\cos\left(t\sqrt{-\lambda}\right) \\
y'(0) &= c_2\sqrt{-\lambda} = 0 \longrightarrow c_2 = 0 \\
y'(2\pi) &= c_1\sqrt{-\lambda}\sinh\left(2\pi\sqrt{-\lambda}\right) = 0 \longrightarrow c_1 = 0
\end{aligned}$$

This is a trivial solution.

The final solution is,

$$\boxed{y = a_0 + \sum_{n=1} a_n \cos\left(\frac{nt}{2}\right)}$$

Example 3

Consider the following problem,

$$\frac{d^2y}{dt^2} + \lambda y = 0, \quad \text{BCs: } y(\pi) = y(-\pi),\ y'(\pi) = y'(-\pi)$$

---

1. **Case** $\lambda > 0$:

$$\begin{aligned} r &= \pm i\sqrt{\lambda} \\ y &= c_1 \cos\left(t\sqrt{\lambda}\right) + c_2 \sin\left(t\sqrt{\lambda}\right) \\ y' &= -c_1\sqrt{\lambda}\sin\left(t\sqrt{\lambda}\right) + c_2\sqrt{\lambda}\cos\left(t\sqrt{\lambda}\right) \end{aligned}$$

From the boundary conditions,

$$\begin{aligned} y(\pi) = y(-\pi) &\longrightarrow 2c_2 \sin\left(\pi\sqrt{\lambda}\right) = 0 \\ y'(\pi) = y'(-\pi) &\longrightarrow 2c_1\sqrt{\lambda}\sin\left(\pi\sqrt{\lambda}\right) = 0 \end{aligned}$$

$$\pi\sqrt{\lambda} = \pi n \longrightarrow \sqrt{\lambda} = n$$

$$\boxed{y_n = a_n \cos(nt) + b_n \sin(nt)}$$

2. **Case** $\lambda = 0$

$$\begin{aligned} y &= c_1 x + c_2 \\ y' &= c_1 \end{aligned}$$

From the boundary conditions,

$$y(\pi) = y(-\pi) \longrightarrow c_1 = 0$$

$$\boxed{y_n = c_n}$$

3. **Case** $\lambda < 0$

$$\begin{aligned}
r &= \pm\sqrt{-\lambda} \\
y &= c_1 \cosh\left(t\sqrt{-\lambda}\right) + c_2 \sinh\left(t\sqrt{-\lambda}\right) \\
y' &= c_1\sqrt{-\lambda}\sinh\left(t\sqrt{-\lambda}\right) + c_2\sqrt{-\lambda}\cosh\left(t\sqrt{-\lambda}\right) \\
y(\pi) = y(-\pi) &\longrightarrow 2c_2 \sinh\left(\pi\sqrt{-\lambda}\right) = 0 \longrightarrow c_2 = 0 \\
y'(\pi) = y'(-\pi) &\longrightarrow 2c_1\sqrt{-\lambda}\sinh\left(\pi\sqrt{-\lambda}\right) = 0 \longrightarrow c_1 = 0
\end{aligned}$$

This yields the trivial solution.

The final solution is,

$$\boxed{y = a_0 + \sum_{n=1}\Big[a_n \cos(nt) + b_n \sin(nt)\Big]}$$

1.3 Variation of Parameters

Here is the general form of the differential equation that we aim to solve:

$$y''(x) + p(x)y'(x) + q(x)y(x) = r(x)$$

Start from the general solution to the homogeneous ODE,

$$y_h = c_1y_1 + c_2y_2$$

Now, we vary the parameters c_1 and c_2, making them functions of x, in order to find the particular solution.

$$\begin{aligned}
y_p &= u(x)y_1(x) + v(x)y_2(x) \\
y_p' &= uy_1' + vy_2' + u'y_1 + v'y_2
\end{aligned}$$

Without loss of generality, we can impose the condition that $\boxed{u'y_1 + v'y_2 = 0}$.

The equation for y_p' becomes simplified with this condition, and it is easier to calculate its derivative.

$$
\begin{aligned}
y_p' &= uy_1' + vy_2' \\
y_p'' &= u'y_1' + uy_1'' + v'y_2' + vy_2''
\end{aligned}
$$

Plug back into the differential equation,

$$
\begin{aligned}
\left(u'y_1' + uy_1'' + v'y_2' + vy_2''\right) + p\left(uy_1' + vy_2'\right) + q\left(u(x)y_1(x) + v(x)y_2(x)\right) &= r \\
u'y_1' + v'y_2' + u\left(y_1'' + py_1' + qy_1\right) + v\left(y_2'' + py_2' + qy_2\right) &= r \\
\boxed{u'y_1' + v'y_2' = r}
\end{aligned}
$$

Therefore, we have the system of equations,

$$
\begin{Bmatrix}
u'y_1 + v'y_2 = 0 \\
u'y_1' + v'y_2' = r
\end{Bmatrix}
$$

$$
\begin{bmatrix} y_1 & y_2 \\ y_1' & y_2' \end{bmatrix}
\begin{bmatrix} u' \\ v' \end{bmatrix}
=
\begin{bmatrix} 0 \\ r \end{bmatrix}
$$

Using Cramer's rule,

$$
W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1y_2' - y_2y_1'
$$

$$
u' = \frac{1}{W}\begin{vmatrix} 0 & y_2 \\ r & y_2' \end{vmatrix} = -\frac{r\,y_2}{W}
$$

$$
v' = \frac{1}{W}\begin{vmatrix} y_1 & 0 \\ y_1' & r \end{vmatrix} = \frac{r\,y_1}{W}
$$

Chapter 2

Fourier Series and Orthogonality

Let's talk about orthogonality, a concept that you have probably heard before in linear algebra. Two vectors $\mathbf{A}$ and $\mathbf{B}$ are said to be orthogonal to each other when their dot product is 0, $\mathbf{A}\cdot\mathbf{B} = 0$. We can represent a vector as a linear combination of orthogonal basis vectors (that do not need to be unit length). For example,

$$\mathbf{A} = a_1\mathbf{u}_1 + a_2\mathbf{u}_2 + a_3\mathbf{u}_3$$

We can find the orthogonal components by taking the dot product with the appropriate basis vector,

$$a_1 = \frac{\mathbf{A}\cdot\mathbf{u}_1}{\mathbf{u}_1\cdot\mathbf{u}_1}, \quad a_2 = \frac{\mathbf{A}\cdot\mathbf{u}_2}{\mathbf{u}_2\cdot\mathbf{u}_2}, \quad a_3 = \frac{\mathbf{A}\cdot\mathbf{u}_3}{\mathbf{u}_3\cdot\mathbf{u}_3}$$

The beauty of linear algebra is that we can generalize this to functions as well, by treating functions as infinite-dimensional vectors. Fourier came up with the idea that we can represent a function as a sum of trigonometric functions. For example,

$$f(x) = a_1 \sin\left(\frac{\pi x}{L}\right) + a_2 \sin\left(\frac{2\pi x}{L}\right) + a_3 \sin\left(\frac{3\pi x}{L}\right) + ...$$

In this case, the dot product is generalized to an **inner product**, defined as,

$$\langle A(x), B(x)\rangle = \int_0^L A(x)B(x)\,dx$$

So that the components are found as,

$$a_1 = \frac{\left\langle f(x), \sin\left(\frac{\pi x}{L}\right)\right\rangle}{\left\langle \sin\left(\frac{\pi x}{L}\right), \sin\left(\frac{\pi x}{L}\right)\right\rangle}, \quad a_2 = \frac{\left\langle f(x), \sin\left(\frac{2\pi x}{L}\right)\right\rangle}{\left\langle \sin\left(\frac{2\pi x}{L}\right), \sin\left(\frac{2\pi x}{L}\right)\right\rangle}, \quad ...$$

We can show that sines and cosines form an orthogonal set of basis functions. The **orthogonality relations** are,

Orthogonality Relations

For $0 < x < L$,

$$\int_0^L \sin\left(\frac{n\pi x}{L}\right)\sin\left(\frac{m\pi x}{L}\right)\,dx = \begin{cases} 0, & n \neq m, \\ \frac{L}{2}, & n = m \end{cases}$$

$$\int_0^L \cos\left(\frac{n\pi x}{L}\right)\cos\left(\frac{m\pi x}{L}\right)\,dx = \begin{cases} 0, & n \neq m, \\ \frac{L}{2}, & n = m \neq 0, \\ L, & n = m = 0 \end{cases}$$

For $-L < x < L$,

$$\int_{-L}^L \sin\left(\frac{n\pi x}{L}\right)\sin\left(\frac{m\pi x}{L}\right)\,dx = \begin{cases} 0, & n \neq m, \\ L, & n = m \end{cases}$$

$$\int_{-L}^L \cos\left(\frac{n\pi x}{L}\right)\cos\left(\frac{m\pi x}{L}\right)\,dx = \begin{cases} 0, & n \neq m, \\ L, & n = m \neq 0, \\ 2L, & n = m = 0 \end{cases}$$

$$\int_{-L}^L \sin\left(\frac{n\pi x}{L}\right)\cos\left(\frac{m\pi x}{L}\right)\,dx = 0$$

Chapter 3

Separation of Variables

3.1 Heat Equation

Example 1: Heat Equation

Let's start off with a simple PDE, the heat equation,

$$\begin{aligned} \frac{\partial u}{\partial t} &= \alpha^2 \frac{\partial^2 u}{\partial x^2} \\ \text{BCs: } u(0,t) &= 0 \\ u(1,t) &= 0 \\ \text{IC: } u(x,0) &= f(x) \end{aligned}$$

Let us assume that the solution can be written as,

$$u(x,t) = \phi(x)T(t)$$

Plugging this into the PDE,

$$\begin{aligned} \phi(x)T'(t) &= \alpha^2 \phi''(x)T(t) \\ \frac{T'(t)}{\alpha^2 T(t)} &= \frac{\phi''(x)}{\phi(x)} = -\lambda \end{aligned}$$

It is customary to let $\lambda = \omega^2$,

$$\begin{cases} \dfrac{dT}{dt} + \omega^2\alpha^2 T(t) = 0 \\ \dfrac{d^2\phi}{dx^2} + \omega^2\phi(x) = 0 \end{cases}$$

The solutions to both ODEs are,

$$\begin{cases} T(t) = c_1 e^{-\omega^2\alpha^2 t} \\ \phi(x) = c_2\cos(\omega x) + c_3\sin(\omega x) \end{cases}$$

From the boundary conditions,

$$\begin{aligned} u(0,t) = \phi(0)T(t) = 0 \quad &\longrightarrow \quad \phi(0) = 0 \quad \longrightarrow \quad c_1 = 0 \\ u(1,t) = \phi(1)T(t) = 0 \quad &\longrightarrow \quad \phi(1) = 0 \quad \longrightarrow \quad c_2\sin(\omega) = 0 \end{aligned}$$

The second boundary condition, $\phi(1) = 0$ is satisfied when $\omega = \pi n$

The eigenfunctions for the solution space are,

$$\begin{aligned} \phi_n(x) &= \sin(\pi n x) \\ T_n(t) &= e^{-(\pi n\alpha)^2 t} \end{aligned}$$

Thus, the final solution is,

$$\boxed{u(x,t) = \sum_n T_n(t)\phi_n(x) = \sum_n A_n e^{-(\pi n\alpha)^2 t}\sin(\pi n x)}$$

To satisfy the initial condition, we simply use the orthogonality properties that we described in Chapter 2.

$$u(x,0) = f(x) = \sum A_n\sin(\pi n x)$$

Multiply both sides by $\sin(\pi m x)$ and integrate,

$$\int_0^1 \sum_n A_n \sin(\pi n x)\sin(\pi m x)\,dx = \int_0^1 f(x)\sin(\pi m x)\,dx$$

$$A_m \int_0^1 \sin^2(\pi m x)dx = \int_0^1 f(x)\sin(\pi m x)\,dx$$

$$A_m = \frac{\int_0^1 f(x)\sin(\pi m x)\,dx}{\int_0^1 \sin^2(\pi m x)dx}$$

Change the subscript back to n and evaluate the integral,

$$\boxed{A_n = 2\int_0^1 f(x)\sin(\pi n x)\,dx}$$

Just for fun, let's try some functions for the initial condition. Let's say that,

$$f(x) = 1$$

$$A_n = 2\int_0^1 \sin(\pi n x)dx = -\frac{2}{\pi n}\cos(\pi n x)\Big|_0^1 = -\frac{2}{\pi n}\left[\cos(\pi n) - 1\right]$$

Solving for the coefficients, $A_1, A_2, A_3, ...,$

$$A_1 = \frac{4}{\pi}, \qquad A_2 = 0, \qquad A_3 = \frac{4}{3\pi},$$
$$A_4 = 0, \qquad A_5 = \frac{4}{5\pi}, \qquad A_6 = 0, \qquad A_7 = \frac{4}{7\pi}.$$

$$\boxed{u(x,t) = \frac{4}{\pi}\left[e^{-(\pi\alpha)^2 t}\sin(\pi x) + e^{-(3\pi\alpha)^2 t}\frac{\sin(3\pi x)}{3} + e^{-(5\pi\alpha)^2 t}\frac{\sin(5\pi x)}{5} + e^{-(7\pi\alpha)^2}\frac{\sin(7\pi x)}{7} + ...\right]}$$

Let's say that,

$$f(x) = \sin(2\pi x) + \frac{1}{3}\sin(4\pi x) + \frac{1}{5}\sin(6\pi x)$$

Then we have,

$$\boxed{u(x,t) = e^{-(2\pi)^2 t}\sin(2\pi x) + \frac{1}{3}e^{-(4\pi)^2 t}\sin(4\pi x) + \frac{1}{5}e^{-(6\pi)^2 t}\sin(6\pi x)}$$

What if the initial condition is,

$$f(x) = x - x^2$$

$$\begin{aligned}
A_n &= 2\int_0^1 f(x)\sin(\pi n x)\,dx = 2\int_0^1 (x-x^2)\sin(\pi n x)\,dx \\
&= 2\left[(x-x^2)\left(-\frac{\cos(\pi n x)}{\pi n}\right) + (1-2x)\left(\frac{\sin(\pi n x)}{\pi^2 n^2}\right) - \frac{2\cos(\pi n x)}{\pi^3 n^3}\right]_0^1 \\
&= -4\left[\frac{(-1)^n}{\pi^3 n^3} - \frac{1}{\pi^3 n^3}\right]
\end{aligned}$$

Then,

$$\boxed{u(x,t) = \frac{8}{\pi^3}\left[e^{-\pi^2 t}\sin(\pi x) + e^{-(3\pi)^2 t}\frac{\sin(3\pi x)}{3^3} + e^{-(5\pi)^2 t}\frac{\sin(5\pi x)}{5^3} + ...\right]}$$

3.2 Wave Equation

Example 2: Wave Equation

$$\begin{aligned}
\frac{\partial^2 u}{\partial t^2} &= c^2\frac{\partial^2 u}{\partial x^2}, \quad 0 < x < L, \quad 0 < t < \infty \\
\text{BCs: } u(0,t) &= 0 \\
u(L,t) &= 0 \\
\text{IC: } u(x,0) &= f(x) \\
\frac{\partial u}{\partial t}(x,0) &= g(x)
\end{aligned}$$

Let us begin the separation of variables technique.

$$
\begin{aligned}
u(x,t) &= \phi(x)T(t) \\
\phi(x)T''(t) &= c^2\phi''(x)T(t) \\
\frac{T''(t)}{c^2T(t)} &= \frac{\phi''(x)}{\phi(x)} = -\lambda
\end{aligned}
$$

This leads to the following system of ODEs,

$$
\begin{cases}
\dfrac{d^2\phi}{dx^2} + \lambda\phi = 0, \quad \phi(0) = 0, \quad \phi(L) = 0 \\
\dfrac{d^2T}{dt^2} + \lambda c^2 T = 0
\end{cases}
$$

For $\lambda > 0$, the solutions to each ODEs are,

$$
\begin{cases}
\phi(x) = c_1\cos\left(x\sqrt{\lambda}\right) + c_2\sin\left(x\sqrt{\lambda}\right) \\
T(t) = c_3\cos\left(ct\sqrt{\lambda}\right) + c_4\sin\left(ct\sqrt{\lambda}\right)
\end{cases}
$$

From $\phi(0) = 0$, we get $c_1 = 0$. From $\phi(L) = 0$, we get $\sqrt{\lambda} = \pi n/L$.

The product solutions are,

$$
\sin\left(\frac{\pi n x}{L}\right)\cos\left(\frac{\pi n c t}{L}\right), \quad \sin\left(\frac{\pi n x}{L}\right)\sin\left(\frac{\pi n c t}{L}\right)
$$

$$
\boxed{u(x,t) = \sum_{n=1}^{\infty}\left[a_n\sin\left(\frac{c\pi n t}{L}\right) + b_n\cos\left(\frac{c\pi n t}{L}\right)\right]\sin\left(\frac{\pi n x}{L}\right)}
$$

From the initial conditions,

$$
\begin{aligned}
u(x,0) &= \sum_{n=1}^{\infty} b_n \sin\left(\frac{\pi n x}{L}\right) = f(x) \\
\frac{\partial u}{\partial t}(x,0) &= \sum_{n=1}^{\infty} a_n \frac{c\pi n}{L}\sin\left(\frac{\pi n x}{L}\right) = g(x)
\end{aligned}
$$

Using orthogonality rules, I can solve for the coefficients,

$$\sum_n \int_0^L b_n \sin\left(\frac{\pi n x}{L}\right) \sin\left(\frac{\pi m x}{L}\right)\, dx = \int_0^L f(x) \sin\left(\frac{\pi m x}{L}\right)\, dx$$
$$b_m \int_0^L \sin^2\left(\frac{\pi m x}{L}\right)\, dx = \int_0^L f(x) \sin\left(\frac{\pi m x}{L}\right)\, dx$$

Change index from m to n,

$$\boxed{b_n = \frac{\int_0^L f(x)\sin\left(\frac{\pi n x}{L}\right)\, dx}{\int_0^L \sin^2\left(\frac{\pi n x}{L}\right)\, dx} = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{\pi n x}{L}\right)\, dx}$$

$$\sum_n \int_0^L a_n \frac{c\pi n}{L} \sin\left(\frac{\pi n x}{L}\right) \sin\left(\frac{\pi m x}{L}\right)\, dx = \int_0^L g(x) \sin\left(\frac{\pi m x}{L}\right)\, dx$$

$$\boxed{a_n = \frac{2}{c\pi n}\int_0^L g(x)\sin\left(\frac{\pi n x}{L}\right)\, dx}$$

2D Wave Equation: Rectangular Region

$$\frac{\partial^2 u}{\partial t^2} = c^2\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right), \quad 0 \le x \le L,\ 0 \le y \le H$$
$$\text{BCs: } u(0,y,t) = 0, \quad u(x,0,t) = 0$$
$$u(L,y,t) = 0, \quad u(x,H,t) = 0$$
$$\text{ICs: } u(x,y,0) = \alpha(x,y)$$
$$\frac{\partial u}{\partial t}(x,y,0) = \beta(x,y)$$

Let's begin the separation of variables process.

$$u(x,y,t) = T(t)U(x,y)$$
$$T''U = c^2\left(T\frac{\partial^2 U}{\partial x^2} + T\frac{\partial^2 U}{\partial y^2}\right)$$
$$\frac{T''}{c^2 T} = \frac{1}{U}\left[\frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2}\right] = -\lambda$$

Here's the system of equations this forms,

$$\begin{cases} T'' + \lambda c^2 T = 0 \\ \dfrac{\partial^2 U}{\partial x^2} + \dfrac{\partial^2 U}{\partial y^2} + \lambda U = 0 \end{cases}$$

with the following boundary conditions,

$$U(0,y) = 0, \quad U(L,y) = 0$$
$$U(x,0) = 0, \quad U(x,H) = 0$$

I focus on the second equation and perform separation of variables again,

$$\begin{aligned} U(x,y) &= X(x)Y(y) \\ X''Y + XY'' + \lambda XY &= 0 \\ \frac{X''}{X} &= -\frac{Y''}{Y} - \lambda = -\mu \end{aligned}$$

We have now reduced the problem to 3 ODEs,

$$\left\{ \begin{aligned} T'' + \lambda c^2 T &= 0 \\ X'' + \mu X &= 0, \quad X(0) = 0,\ X(L) = 0 \\ Y'' + (\lambda - \mu) Y &= 0, \quad Y(0) = 0,\ Y(H) = 0 \end{aligned} \right\}$$

The eigenfunctions of the second equation are,

$$X_n = \sin\left(\frac{\pi n x}{L}\right), \quad \text{corresponding to } \mu_n = \left(\frac{\pi n}{L}\right)^2$$

For each μ_n, there is a corresponding λ_{nm},

$$Y_{nm} = \sin\left(\frac{\pi m y}{H}\right), \quad \text{corresponding to } \lambda_{nm} - \mu_n = \left(\frac{\pi m}{H}\right)^2$$

So,m now the spatial part of the solution is,

$$\boxed{U_{nm}(x,y) = X_n(x) Y_{nm}(y) = \sin\left(\frac{\pi n x}{L}\right) \sin\left(\frac{\pi m y}{H}\right)}$$

Going back to the equation for T,

$$T = A\cos\left(ct\sqrt{\lambda}\right) + B\sin\left(ct\sqrt{\lambda}\right)$$

$$\boxed{T_{nm} = A_{nm}\cos\left(ct\sqrt{\lambda_{nm}}\right) + B_{nm}\sin\left(ct\sqrt{\lambda_{nm}}\right)}$$

where,

$$\lambda_{nm} = \mu_n + \left(\frac{\pi m}{H}\right)^2 = \left(\frac{\pi n}{L}\right)^2 + \left(\frac{\pi m}{H}\right)^2$$

Finally,

$$u = \sum_m \sum_n U_{nm} T_{nm}$$

$$\boxed{\begin{aligned} u = &\sum_{m=1}\sum_{n=1} A_{nm}\sin\left(\frac{\pi n x}{L}\right)\sin\left(\frac{\pi m y}{H}\right)\cos\left(ct\sqrt{\lambda_{nm}}\right) \\ &+ \sum_{m=1}\sum_{n=1} B_{nm}\sin\left(\frac{\pi n x}{L}\right)\sin\left(\frac{\pi m y}{H}\right)\sin\left(ct\sqrt{\lambda_{nm}}\right) \end{aligned}}$$

Let's plug this into the first IC,

$$u(x,y,0) = \sum_{m=1}\left[\sum_{n=1} A_{nm}\sin\left(\frac{\pi n x}{L}\right)\right]\sin\left(\frac{\pi m y}{H}\right) = \alpha(x,y)$$

We can now regard the term in brackets as the coefficient of a Fourier sine series in y of $\alpha(x,y)$.

$$\sum_{n=1} A_{nm}\sin\left(\frac{\pi n x}{L}\right) = \frac{2}{H}\int_0^H \alpha(x,y)\sin\left(\frac{\pi m y}{H}\right)\,dy$$

We now regard A_{nm} as the coefficient of a Fourier sine series of the right hand side,

$$A_{nm} = \frac{2}{L}\int_0^L \left[\frac{2}{H}\int_0^H \alpha(x,y)\sin\left(\frac{\pi m y}{H}\right)\,dy\right]\sin\left(\frac{\pi n x}{L}\right)\,dx$$

$$\boxed{A_{nm} = \frac{4}{LH}\int_0^L\int_0^H \alpha(x,y)\sin\left(\frac{\pi m y}{H}\right)\sin\left(\frac{\pi n x}{L}\right)\,dy\,dx}$$

We follow the same process for the second IC,

$$\sum_{m=1}\left[\sum_{n=1} B_{nm} c\sqrt{\lambda_{nm}} \sin\left(\frac{\pi n x}{L}\right)\right] \sin\left(\frac{\pi m y}{H}\right) = \beta(x,y)$$
$$\sum_{n=1} B_{nm} c\sqrt{\lambda_{nm}} \sin\left(\frac{\pi n x}{L}\right) = \frac{2}{H}\int_0^H \beta(x,y)\sin\left(\frac{\pi m y}{H}\right)\, dy$$
$$B_{nm} c\sqrt{\lambda_{nm}} = \frac{2}{L}\int_0^L \left[\frac{2}{H}\int_0^H \beta(x,y)\sin\left(\frac{\pi m y}{H}\right)\, dy\right]\sin\left(\frac{\pi n x}{L}\right)\, dx$$

$$\boxed{B_{nm} = \frac{4}{LHc\sqrt{\lambda_{nm}}}\int_0^L\int_0^H \beta(x,y)\sin\left(\frac{\pi m y}{H}\right)\sin\left(\frac{\pi n x}{L}\right)\, dy\, dx}$$

3.3 Laplace's Equation: Interior Dirichlet Boundary Conditions for Circle

Laplace's Equation: Interior Dirichlet Boundary Conditions for Circle

$$\nabla^2 u = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2} = 0, \quad 0 < r < 1,\ 0 < \theta < 2\pi$$
$$\text{BC: } u(1,\theta) = g(\theta)$$

Use separation of variables, $u(r,\theta) = R(r)\Theta(\theta)$,

$$R''\Theta + \frac{1}{r}R'\Theta + \frac{1}{r^2}R\Theta'' = 0$$
$$\frac{R''}{R} + \frac{1}{r}\frac{R'}{R} + \frac{1}{r^2}\frac{\Theta''}{\Theta} = 0$$
$$r^2\frac{R''}{R} + r\frac{R'}{R} = -\frac{\Theta''}{\Theta} = \lambda^2$$

I get a system of equations,

$$\left\{\begin{aligned} \Theta'' + \lambda^2\Theta = 0 \\ r^2R'' + rR' - R\lambda^2 = 0 \end{aligned}\right\}$$

Let's focus on the equation for R,

<u>For $\lambda = 0$</u>

$$r^2R'' + rR' = 0$$

Let $w = R'$,

$$\begin{aligned} r^2w' + rw &= 0 \\ w' + \frac{w}{r} &= 0 \\ \frac{w'}{w} &= -\frac{1}{r} \\ \ln(w) &= -\ln(r) + C \\ w &= C(1/r) = R' \\ R &= C\ln(r) + D \end{aligned}$$

For $\lambda > 0$

Guess $R = r^m$.

$$\begin{aligned} m(m-1)r^m + mr^m - r^m\lambda^2 &= 0 \\ r^m\left[m(m-1) + m - \lambda^2\right] &= 0 \\ m &= \pm\lambda \end{aligned}$$

Therefore,

$$R = Ar^\lambda + Br^{-\lambda}$$

We throw out $\ln(r)$ and $r^{-\lambda}$ since they become unbounded as $r \to 0$,

$$\boxed{\begin{array}{l} \lambda \geq 0 \\ R = Ar^\lambda \\ \Theta = A\cos(\lambda\theta) + B\sin(\lambda\theta) \end{array}}$$

Thus, the solution is,

$$\boxed{\begin{aligned} u &= \sum_{n=0}^{\infty} r^n \left[a_n \cos(n\theta) + b_n \sin(n\theta)\right] \\ a_0 &= \frac{1}{2\pi}\int_0^{2\pi} g(\theta)\, d\theta \\ a_n &= \frac{1}{\pi}\int_0^{2\pi} g(\theta)\cos(n\theta)\, d\theta \\ b_n &= \frac{1}{\pi}\int_0^{2\pi} g(\theta)\sin(n\theta)\, d\theta \end{aligned}}$$

3.4 Eigenfunction Expansion

$$\begin{aligned} \frac{\partial u}{\partial t} &= \alpha^2 \frac{\partial^2 u}{\partial x^2} + f(x,t), \quad 0 < x < 1 \\ \text{BCs: } u(0,t) &= 0 \\ u(1,t) &= 0 \\ \text{IC: } u(x,0) &= \phi(x) \end{aligned}$$

Let's decompose $f(x,t)$ first,

$$f(x,t) = \sum_n f_n(t) X_n(x)$$

where $X_n(x)$ are the associated eigenfunctions of the following system,

$$X''(x) + \lambda^2 X = 0, \quad X(0) = 0,\ X(1) = 0$$

the solution of which is,

$$X_n = \sin(\lambda x), \quad \text{where } \lambda = \pi n$$

The expansion for $f(x,t)$ becomes,

$$\boxed{f(x,t) = \sum_n f_n(t)\sin(\pi n x)}$$

The coefficients $f_n(t)$ are obtained using orthogonality properties,

$$\begin{aligned}\int_0^1 f(x,t)\sin(\pi m x)\,dx &= \int_0^1 \sum_n f_n(t)\sin(\pi n x)\sin(\pi m x)\,dx \\ &= f_m(t)\int_0^1 \sin^2(\pi m x)\,dx\end{aligned}$$

Solve for f_m and switch the m to n index,

$$\boxed{f_n(t) = \frac{\int_0^1 f(x,t)\sin(\pi n x)\,dx}{\int_0^1 \sin^2(\pi n x)\,dx} = 2\int_0^1 f(x,t)\sin(\pi n x)\,dx}$$

According to the theory of eigenfunction expansion, we can write the solution to the general PDE as,

$$u(x,t) = \sum_n T_n(t)\sin(\pi n x)$$

Plug this into the PDE,

$$\sum_n T_n'(t)\sin(\pi n x) = -\alpha^2 \sum_n T_n(t)(\pi n)^2 \sin(\pi n x) + \sum_n f_n(t)\sin(\pi n x)$$

$$\boxed{\sum_n \left[T_n'(t) + (\alpha\pi n)^2 T_n(t) - f_n(t)\right]\sin(\pi n x) = 0}$$

Plug into boundary condition,

$$u(x,0) = \sum_n T_n(0)\sin(\pi n x) = \phi(x)$$

$$\boxed{T_n(0) = 2\int_0^1 \phi(x)\sin(\pi n x)\,dx = a_n}$$

The ODE is T_n is,

$$\boxed{\begin{aligned} T_n'(t) + (\alpha\pi n)^2 T_n - f_n(t) = 0 \\ T_n(0) = a_n\end{aligned}}$$

If I multiply by an integrating factor $e^{(\alpha\pi n)^2 t}$,

$$\begin{aligned}\left[T_n e^{(\alpha\pi n)^2 t}\right]' &= f_n(t)e^{(\alpha\pi n)^2 t} \\ T_n &= e^{-(\alpha\pi n)^2 t}\int f_n e^{(\alpha\pi n)^2 t}\,dt + c e^{-(\alpha\pi n)^2 t}\end{aligned}$$

Using $T_n(0) = a_n = c$,

$$T_n = \int_0^t f_n(\tau) e^{-(\alpha\pi n)^2(t-\tau)}\, d\tau + a_n e^{-(\alpha\pi n)^2 t}$$

The final solution is,

$$\boxed{u(x,t) = \underbrace{\sum_n \sin(\pi n x) \int_0^t f_n(\tau) e^{-(\alpha\pi n)^2(t-\tau)}\, d\tau}_{\text{steady term}} + \underbrace{\sum_n a_n e^{-(\alpha\pi n)^2 t} \sin(\pi n x)}_{\text{transient term}}}$$

Chapter 4

Sturm-Liouville Problem

The following is known as the Sturm-Liouville differential equation,

$$\boxed{\frac{d}{dx}\left[p(x)\frac{d\phi}{dx}\right] + q(x)\phi + \lambda\sigma(x)\phi = 0}$$

It is useful to define an operator,

$$L(\;) = \frac{d}{dx}\left[p\frac{d}{dx}(\;)\right] + q\cdot(\;)$$

So that the Sturm-Liouville problem becomes, in simpler notation,

$$L(\phi) = -\lambda\sigma\phi$$

ϕ is an eigenfunction with corresponding eigenvalue λ. The eigenfunctions are othogonal to one another.

$$\int_a^b \phi_n\phi_m\sigma\, dx = 0, \quad (m \neq n)$$

Any function can be represented as a linear combination of the eigenfunctions,

$$\boxed{f(x) = \sum_{n=1}^{\infty} a_n\phi_n(x)}$$

The coefficients a_n are evaluated with the following process,

$$\begin{aligned}
\int_a^b f(x)\phi_m\sigma\,dx &= \int_a^b \sum_{n=1}^{\infty} a_n\phi_n\phi_m\sigma\,dx \\
&= \sum_{n=1}^{\infty} a_n \int_a^b \phi_n\phi_m\sigma\,dx \\
&= a_m \int_a^b \phi_m^2\sigma\,dx
\end{aligned}$$

Switching subscripts from m to n and solving for a_n

$$\boxed{a_n = \frac{\int_a^b f(x)\phi_n(x)\sigma(x)\,dx}{\int_a^b \phi_n^2(x)\sigma(x)\,dx}}$$

Lagrange's Identity (or Green's Formula) will be important as well.

$$\boxed{\begin{aligned}
uL(v) - vL(u) &= \frac{d}{dx}\left[p\left(u\frac{dv}{dx} - v\frac{du}{dx}\right)\right] \\
\int_a^b \left[uL(v) - vL(u)\right]\,dx &= p\left(u\frac{dv}{dx} - v\frac{du}{dx}\right)\Bigg|_{x=a}^{x=b}
\end{aligned}}$$

4.1 Wave Equation for Circular Membrane

Wave Equation for Circular Region

Let's try to solve the wave equation for a circular region.

$$\begin{aligned}
\frac{\partial^2 u}{\partial t^2} &= c^2\nabla^2 u \\
\text{BC: } u(a,\theta,t) &= 0 \\
\text{IC: } u(r,\theta,0) &= \alpha(r,\theta) \\
\frac{\partial u}{\partial t}(r,\theta,0) &= \beta(r,\theta)
\end{aligned}$$

Let's do separation of variables,

$$
\begin{aligned}
u(r,\theta,t) &= \phi(r,\theta)T(t) \\
\phi T'' &= c^2 T \nabla^2 \phi \\
\frac{T''}{c^2 T} &= \frac{\nabla^2\phi}{\phi} = -\lambda
\end{aligned}
$$

$$
\begin{Bmatrix}
T'' + \lambda c^2 T = 0 \\
\nabla^2\phi + \lambda\phi = 0, \quad \phi(a,\theta) = 0
\end{Bmatrix}
$$

Let's do separation of variables again for the second equation above,

$$\phi(r,\theta) = R(r)\Theta(\theta)$$

Plug in the second equation,

$$
\begin{aligned}
\frac{1}{r}\frac{\partial}{\partial r}\left[r\frac{\partial\phi}{\partial r}\right] + \frac{1}{r^2}\frac{\partial^2\phi}{\partial\theta^2} + \lambda\phi = 0 \\
\frac{\Theta}{r}\frac{\partial}{\partial r}\left(rR'\right) + \frac{R\Theta''}{r^2} + \lambda R\Theta = 0 \\
\frac{r}{R}\frac{\partial}{\partial r}\left(rR'\right) + \lambda r^2 = -\frac{\Theta''}{\Theta} = \mu
\end{aligned}
$$

We get a system of ODEs,

$$
\begin{Bmatrix}
T'' + \lambda c^2 T = 0 \\
\Theta'' + \mu\Theta = 0, \quad \Theta(-\pi) = \Theta(\pi),\ \Theta'(-\pi) = \Theta'(\pi) \\
r\frac{d}{dr}\left[r\frac{dR}{dr}\right] + R(\lambda r^2 - \mu) = 0, \quad R(a) = 0,\ |R(0)| < \infty
\end{Bmatrix}
$$

Let us focus on the equation for Θ,

$$
\begin{aligned}
&\mu > 0 \\
&\Theta = A\cos\left(\theta\sqrt{\mu}\right) + B\sin\left(\theta\sqrt{\mu}\right)
\end{aligned}
$$

From the boundary conditions, we conclude that $\mu = m^2$, where m is an integer.

The R differential equation can be put into Sturm-Liouville form by dividing by r,

$$\frac{d}{dr}\left[r\frac{dR}{dr}\right] + R\left(\lambda r - \frac{m^2}{r}\right) = 0$$

By expanding out the derivative, we get an easier form of the equation,

$$\boxed{r^2\frac{d^2R}{dr^2} + r\frac{dR}{dr} + R\left(\lambda r^2 - m^2\right) = 0}$$

Using the simple scaling transformation, $z = r\sqrt{\lambda}$,

$$\boxed{z^2\frac{d^2R}{dz^2} + z\frac{dR}{dz} + (z^2 - m^2)R = 0}$$

The above equation is a very special type of differential equation known as **Bessel's differential equation**. The solution cannot be expressed in elementary functions.

$$\begin{aligned} R(z) &= c_1 J_m(z) + c_2 Y_m(z) \\ R &= c_1 J_m\left(r\sqrt{\lambda}\right) + c_2 Y_m\left(r\sqrt{\lambda}\right) \end{aligned}$$

We know that $c_2 = 0$, since $Y_m(r\sqrt{\lambda})$ is unbounded as $r \to 0$.

$$R = c_1 J_m\left(r\sqrt{\lambda}\right)$$

From the condition that $R(r = a) = 0$,

$$\begin{aligned} J_m\left(a\sqrt{\lambda}\right) &= 0 \\ a\sqrt{\lambda} &= z_{mn}, \quad \text{where } z_{mn} \text{ is the nth zero of } J_m(z) \\ &\boxed{\lambda_{mn} = \left(\frac{z_{mn}}{a}\right)^2} \end{aligned}$$

To summarize, the solution to the system of ODES is,

$$\boxed{\begin{aligned} T_{mn} &= C_1\cos\left(ct\sqrt{\lambda_{mn}}\right) + C_2\sin\left(ct\sqrt{\lambda_{mn}}\right) \\ \Theta_m &= C_3\cos(m\theta) + C_4\sin(m\theta) \\ R_{mn} &= C_5 J_m\left(r\sqrt{\lambda_{mn}}\right) + C_6 Y_m\left(r\sqrt{\lambda_{mn}}\right) \end{aligned}}$$

Since $u(r,\theta,t) = R(r)\Theta(\theta)T(t)$, the product solutions are (there are four product terms),

$$J_m\left(r\sqrt{\lambda_{mn}}\right)\begin{Bmatrix}\cos(m\theta)\\ \sin(m\theta)\end{Bmatrix}\begin{Bmatrix}\cos\left(ct\sqrt{\lambda_{mn}}\right)\\ \sin\left(ct\sqrt{\lambda_{mn}}\right)\end{Bmatrix}$$

If $\beta(r,\theta) = 0$, the problem is much simpler since there are two product terms now.

$$\boxed{\begin{aligned} u(r,\theta,t) = &\sum_{m=0}^{\infty}\sum_{n=1}^{\infty} A_{mn}J_m\left(r\sqrt{\lambda_{mn}}\right)\cos(m\theta)\cos\left(ct\sqrt{\lambda_{mn}}\right)\\ &+\sum_{m=1}^{\infty}\sum_{n=1}^{\infty} B_{mn}J_m\left(r\sqrt{\lambda_{mn}}\right)\sin(m\theta)\cos\left(ct\sqrt{\lambda_{mn}}\right)\end{aligned}}$$

Wave Equation: Circularly Symmetric Case

In the circularly symmetric case, we remove all dependencies on θ, so that $u = u(r,t)$.

$$\begin{aligned} \frac{\partial^2 u}{\partial t^2} &= \frac{c^2}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right), \quad 0 < r < a\\ \text{BC: } u(a,t) &= 0\\ \text{ICs: } u(r,0) &= \alpha(r)\\ \frac{\partial u}{\partial t}(r,0) &= \beta(r) \end{aligned}$$

Let's begin with separation of variables,

$$u(r,t) = \phi(r)T(t)$$

Plug into the PDE,

$$\begin{aligned} \phi(r)T''(t) &= \frac{c^2}{r}\frac{\partial}{\partial r}\left[r\phi'(r)T(t)\right]\\ \frac{T''}{c^2T} &= \frac{1}{r\phi}\frac{\partial}{\partial r}(r\phi') = -\lambda \end{aligned}$$

$$\boxed{\begin{aligned} &T'' + \lambda c^2 T = 0\\ &\frac{d}{dr}\left(r\frac{d\phi}{dr}\right) + \lambda r\phi = 0, \quad \phi(a) = 0,\ |\phi(0)| < \infty \end{aligned}}$$

The equation for ϕ is in Sturm-Liouville form, with a $r-$weighted inner product. If I make the transformation, $z = r\sqrt{\lambda}$,

$$\frac{d}{dz}\left[z\frac{d\phi}{dz}\right] + z\phi = 0, \quad z^2\frac{d^2\phi}{dz^2} + z\frac{d\phi}{dz} + z^2\phi = 0$$

The solution is found using Bessel's functions,

$$\begin{aligned} \phi &= c_1 J_0(z) + c_2 Y_0(z) \\ &= c_1 J_0\left(r\sqrt{\lambda}\right) + c_2 Y_0\left(r\sqrt{\lambda}\right) \end{aligned}$$

We find that $c_2 = 0$ since $\phi(0)$ must be bounded. From the other boundary condition, $\phi(a) = 0$,

$$\begin{aligned} c_1 J_0\left(a\sqrt{\lambda}\right) &= 0 \\ a\sqrt{\lambda_n} &= z_n \\ \lambda_n &= \left(\frac{z_n}{a}\right)^2 \end{aligned}$$

The equation for T is,

$$\boxed{T(t) = A\cos\left(ct\sqrt{\lambda}\right) + B\sin\left(ct\sqrt{\lambda}\right)}$$

$$\boxed{u(r,t) = \sum_{n=1}^{\infty} a_n J_0\left(r\sqrt{\lambda_n}\right)\cos\left(ct\sqrt{\lambda_n}\right) + \sum_{n=1}^{\infty} b_n J_0\left(r\sqrt{\lambda_n}\right)\sin\left(ct\sqrt{\lambda_n}\right)}$$

From the initial condition,

$$u(r,0) = \sum_{n=1}^{\infty} a_n J_0\left(r\sqrt{\lambda_n}\right) = \alpha(r)$$

$$a_n = \frac{\int_0^a r\alpha(r) J_0\left(r\sqrt{\lambda_n}\right)\, dr}{\int_0^a J_0^2\left(r\sqrt{\lambda_n}\right)\, rdr}$$

4.2 Laplace's Equation in a Circular Cylinder

Laplace's Equation in a Circular Cylinder

$$\begin{aligned}\nabla^2 u &= 0\\ \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2}+\frac{\partial^2 u}{\partial z^2} &= 0, \quad 0<r<a,\ 0<z<H\\ u(r,\theta,H) &= \beta(r,\theta)\\ u(r,\theta,0) &= \alpha(r,\theta)\\ u(a,\theta,z) &= \gamma(\theta,z)\end{aligned}$$

Let's begin the separation of variables approach,

$$u(r,\theta,z)=f(r)g(\theta)h(z)$$

Plug into the PDE,

$$\begin{aligned}\frac{1}{r}\frac{\partial}{\partial r}\left\{r\frac{\partial}{\partial r}\Big[f(r)g(\theta)h(z)\Big]\right\}+\frac{1}{r^2}\frac{\partial^2}{\partial \theta^2}\Big[f(r)g(\theta)h(z)\Big]+\frac{\partial^2}{\partial z^2}\Big[f(r)g(\theta)h(z)\Big] &= 0\\ \frac{1}{rf}\frac{\partial}{\partial r}\left[r\frac{df}{dr}\right]+\frac{1}{r^2 g}\frac{\partial^2 g}{\partial\theta^2}+\frac{1}{h}\frac{\partial^2 h}{\partial z^2} &= 0\end{aligned}$$

Set $(1/h)(\partial^2 h/\partial z^2)=\lambda$,

$$\begin{aligned}\frac{r}{f}\frac{d}{dr}\left[r\frac{df}{dr}\right]+\frac{1}{g}\frac{d^2 g}{d\theta^2}+\lambda r^2 &= 0\\ \frac{r}{f}\frac{d}{dr}\left[r\frac{df}{dr}\right]+\lambda r^2 &= -\frac{1}{g}\frac{d^2 g}{d\theta^2}=\mu\end{aligned}$$

Set $\mu=m^2$,

$$\boxed{\begin{aligned}\frac{d^2 g}{d\theta^2}+m^2 g &= 0\\ r\frac{d}{dr}\left[r\frac{df}{dr}\right]+\left(\lambda r^2-m^2\right)f &= 0\\ \frac{d^2 h}{dz^2}-h\lambda &= 0\end{aligned}} \tag{4.1}$$

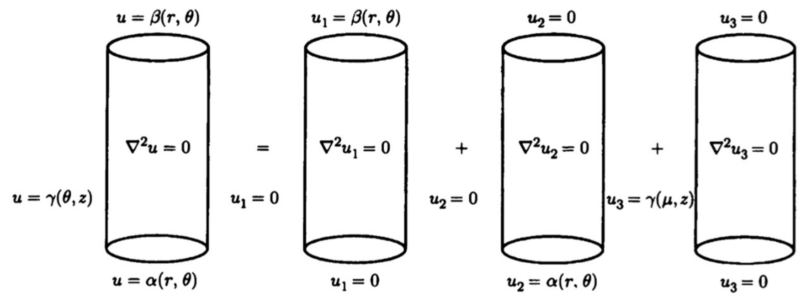

Figure: Laplace's equation in a circular cylinder

The best way to solve this is to break the problem down into three simpler problems, each with two homogenous boundary conditions and one nonhomogenous condition . Let's focus on u_1 first,

$$\begin{aligned} \nabla^2 u_1 &= 0 \\ u_1(r,\theta,0) &= 0 \\ u_1(r,\theta,H) &= \beta(r,\theta) \\ u_1(a,\theta,z) &= 0 \end{aligned}$$

The boundary conditions imply that $h(0) = 0$ and $f(a) = 0$

Since $\lambda > 0$, the solutions to the system of ODES (Eq. 4.1) are,

$$\boxed{\begin{aligned} &g = c_1 \cos(m\theta) + c_2 \sin(m\theta) \\ &f = c_3 J_m\left(r\sqrt{\lambda}\right) + c_4 Y_m\left(r\sqrt{\lambda}\right) \\ &h = c_5 \sinh\left(z\sqrt{\lambda}\right) + c_6 \cosh\left(z\sqrt{\lambda}\right) \end{aligned}}$$

The product solutions are simply,

$$J_m\left(r\sqrt{\lambda_{mn}}\right)\begin{Bmatrix}\cos(m\theta)\\ \sin(m\theta)\end{Bmatrix}\sinh\left(z\sqrt{\lambda_{mn}}\right)$$

And the equation for u_1 is,

$$\boxed{\begin{aligned} u_1 &= \sum_{m=0}^{\infty}\sum_{n=1}^{\infty} A_{mn} J_m\left(r\sqrt{\lambda_{mn}}\right)\cos(m\theta)\sinh\left(z\sqrt{\lambda_{mn}}\right) \\ &+ \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} B_{mn} J_m\left(r\sqrt{\lambda_{mn}}\right)\sin(m\theta)\sinh\left(z\sqrt{\lambda_{mn}}\right) \end{aligned}}$$

For the u_3 problem,

$$\begin{aligned} \nabla^2 u_3 &= 0 \\ u_3(r,\theta,0) &= 0 \\ u_3(r,\theta,H) &= 0 \\ u_3(a,\theta,z) &= \gamma(\theta,z) \end{aligned}$$

The boundary conditions imply that $h(0) = 0$ and $h(H) = 0$.

In this case, we want $h(z)$ to be oscillatory, so λ should be negative. Let $\lambda = -\omega^2$,

$$h = c_1 \sin(\omega z) = c_1 \sin\left(\frac{\pi n z}{H}\right), \quad \lambda = -\left(\frac{\pi n}{H}\right)^2$$

The differential equation for f is,

$$r\frac{d}{dr}\left[r\frac{df}{dr}\right] + \left[-\left(\frac{\pi n}{H}\right)^2 r^2 - m^2\right] f = 0$$

Make the change of variables, $s = i(\pi n/H)r$,

$$\begin{aligned} s\frac{d}{ds}\left[s\frac{df}{ds}\right] + \left(s^2 - m^2\right) f &= 0 \\ s^2\frac{d^2 f}{ds^2} + s\frac{df}{ds} + (s^2 - m^2) f &= 0 \end{aligned}$$

The solution is,

$$\begin{aligned} f &= c_1 J_m(s) + c_2 Y_m(s) \\ &= c_1 J_m\left[i\left(\frac{\pi n}{H}\right) r\right] + c_2 Y_m\left[i\left(\frac{\pi n}{H}\right) r\right] \\ &= c_1 K_m\left(\frac{n\pi}{H} r\right) + c_2 I_m\left(\frac{n\pi}{H} r\right) \end{aligned}$$

$$\boxed{\begin{aligned} u_3(r,\theta,z) = &\sum_{m=0}^{\infty}\sum_{n=1}^{\infty} E_{mn} I_m\left(\frac{n\pi}{H}r\right)\sin\left(\frac{n\pi z}{H}\right)\cos(m\theta) \\ &+ \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} F_{mn} I_m\left(\frac{n\pi}{H}r\right)\sin\left(\frac{n\pi z}{H}\right)\sin(m\theta) \end{aligned}}$$

Chapter 5

Fourier Transform

5.1 Motivation for the Exponential Fourier Transform

Let's motivate the Fourier Transform,

$$\begin{aligned} \frac{\partial u}{\partial t} &= k\frac{\partial^2 u}{\partial x^2}, \quad -\infty < x < \infty \\ \text{IC: } u(x,0) &= f(x) \end{aligned}$$

Let's solve this problem using the method of the previous section: **separation of variables**.

$$\begin{aligned} u(x,t) &= \phi(x)T(t) \\ \phi(x)T'(t) &= k\phi''(x)T(t) \\ \frac{T'(t)}{kT(t)} &= \frac{\phi''(x)}{\phi(x)} = -\lambda \end{aligned}$$

System of ODEs,

$$\begin{cases} \dfrac{dT}{dt} + k\lambda T = 0 \\ \dfrac{d^2\phi}{dx^2} + \lambda\phi = 0, \quad |\phi(\pm\infty)| < \infty \end{cases}$$

The product solutions for $u(x,t)$ are,

$$e^{-k\lambda t}\cos(x\sqrt{\lambda}), \quad e^{-k\lambda t}\sin(x\sqrt{\lambda})$$

The general solution is,

$$\boxed{\begin{aligned} u(x,t) &= \int\limits_0^\infty \left[c_1(\lambda)e^{-\lambda kt}\cos\left(x\sqrt{\lambda}\right) + c_2(\lambda)e^{-\lambda kt}\sin\left(x\sqrt{\lambda}\right)\right]\, d\lambda \\ &= \int\limits_0^\infty \left[A(\omega)e^{-k\omega^2 t}\cos(\omega x) + B(\omega)e^{-k\omega^2 t}\sin(\omega x)\right]\, d\omega \end{aligned}}$$

$$u(x,0) = \int_0^\infty \left[A(\omega)\cos(\omega x) + B(\omega)\sin(\omega x)\right]\, d\omega = f(x)$$

If I use complex numbers, I can considerably simplify the boxed equation,

$$\boxed{u(x,t) = \int\limits_{-\infty}^\infty c(\omega)e^{-i\omega x}e^{-k\omega^2 t}\, d\omega} \tag{5.1}$$

$$f(x) = \int\limits_{-\infty}^\infty c(\omega)e^{-i\omega x}\, d\omega$$

The coefficient is evaluated as,

$$c(\omega) = \frac{1}{2\pi}\int\limits_{-\infty}^\infty f(x)e^{i\omega x}\, dx$$

So that Eq. 5.1 becomes,

$$\begin{aligned} u(x,t) &= \int\limits_{-\infty}^\infty \left[\frac{1}{2\pi}\int\limits_{-\infty}^\infty f(\overline{x})e^{i\omega\overline{x}}\, d\overline{x}\right] e^{-i\omega x}e^{-k\omega^2 t}\, d\omega \\ &= \frac{1}{2\pi}\int\limits_{-\infty}^\infty f(\overline{x})\left[\int_{-\infty}^\infty e^{-k\omega^2 t}e^{-i\omega(x-\overline{x})}\, d\omega\right] d\overline{x} \end{aligned}$$

Let us define the following function,

$$g(x) = \int\limits_{-\infty}^\infty e^{-k\omega^2 t}e^{-i\omega x}\, d\omega = \sqrt{\frac{\pi}{kt}}\exp\left[-\frac{x^2}{4kt}\right]$$

We will later see that this is defined as the **inverse Fourier transform** of $\exp(-k\omega^2 t)$. Using this function, we can rewrite the term in brackets.

$$
\begin{aligned}
u(x,t) &= \frac{1}{2\pi}\int_{-\infty}^{\infty} f(\overline{x})g(x-\overline{x})\,d\overline{x} \\
&= \frac{1}{2\pi}\int_{-\infty}^{\infty} f(\overline{x})\sqrt{\frac{\pi}{kt}}\exp\left[-\frac{(x-\overline{x})}{4kt}\right]d\overline{x}
\end{aligned}
$$

$$
\boxed{u(x,t) = \int_{-\infty}^{\infty} f(\overline{x})\frac{1}{\sqrt{4\pi kt}}\exp\left[-\frac{(x-\overline{x})^2}{4kt}\right]d\overline{x}}
$$

5.2 Exponential Fourier Transform Properties

Here are the definitions of the Fourier Transform,

$$
F(\omega) = \mathscr{F}[f(x)] = \frac{1}{2\pi}\int_{-\infty}^{\infty} f(x)e^{i\omega x}\,dx \quad\Longleftrightarrow\quad f(x) = \mathscr{F}^{-1}[F(\omega)] = \int_{-\infty}^{\infty} F(\omega)e^{-i\omega x}\,d\omega
$$

$f(x)$	$F(\omega)$
$\frac{df}{dx}$	$(-i\omega)F(\omega)$
$\frac{d^2f}{dx^2}$	$(-i\omega)^2F(\omega)$
$(f*g)(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} f(\overline{x})g(x-\overline{x})\,d\overline{x}$	$F(\omega)G(\omega)$
$\sqrt{\frac{\pi}{\alpha}}e^{-\frac{x^2}{4\alpha}}$	$e^{-\alpha\omega^2}$
$f(x-\beta)$	$F(\omega)e^{i\omega\beta}$
$\delta(x-x_0)$	$\frac{1}{2\pi}e^{i\omega x_0}$

5.3 Exponential FT Example

Fourier Transform Example

We solve the same problem as in section 5.1 but with the full power of the Fourier transform.

$$\begin{aligned} \frac{\partial u}{\partial t} &= k\frac{\partial^2 u}{\partial x^2}, \quad -\infty < x < \infty \\ \text{IC: } u(x,0) &= f(x) \end{aligned}$$

The Fourier transform is defined as,

$$\mathscr{F}\left[u(x,t)\right] = U(\omega,t) = \frac{1}{2\pi}\int\limits_{-\infty}^{\infty} u(x,t)e^{i\omega x}\,dx$$

Let's Fourier transform both sides.

$$\mathscr{F}\left[\frac{\partial u}{\partial t}\right] = k\mathscr{F}\left[\frac{\partial^2 u}{\partial x^2}\right]$$

Using the derivative properties of the Fourier transform,

$$\frac{\partial}{\partial t}\left[U(\omega,t)\right] = -k\omega^2 U(\omega,t)$$

The solution to this ODE is,

$$\begin{aligned} U(\omega,t) &= c(\omega)e^{-k\omega^2 t} \\ &= \mathscr{F}\left[f(x)\right]\mathscr{F}\left[\sqrt{\frac{\pi}{kt}}\exp\left(-\frac{x^2}{4kt}\right)\right] \end{aligned}$$

Using the convolution theorem,

$$\boxed{u(x,t) = \frac{1}{2\pi}\int\limits_{-\infty}^{\infty} f(\overline{x})\sqrt{\frac{\pi}{kt}}\exp\left[-\frac{(x-\overline{x})^2}{4kt}\,d\overline{x}\right]}$$

Wave Equation Fourier Transform

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad -\infty < x < \infty$$
$$\text{ICs: } u(x,0) = f(x)$$
$$\frac{\partial u}{\partial t}(x,0) = 0$$

Fourier transform both sides,

$$\mathscr{F}\left[\frac{\partial^2 u}{\partial t^2}\right] = c^2 \mathscr{F}\left[\frac{\partial^2 u}{\partial x^2}\right]$$
$$\frac{\partial^2 U}{\partial t^2} = -c^2\omega^2 U$$

The solution to the above ODE is,

$$U = A\cos(\omega ct) + B\sin(\omega ct)$$

Fourier transform the ICs,

$$U(\omega,0) = \frac{1}{2\pi}\int_{-\infty}^{\infty} f(x)e^{i\omega x}\,dx$$
$$\frac{\partial U}{\partial t}(\omega,0) = 0$$

From the ICs, we get that $A = U(\omega,0)$ and $B = 0$,

$$U(\omega,t) = U(\omega,0)\cos(\omega ct)$$
$$u(x,t) = \int_{-\infty}^{\infty} U(\omega,0)\cos(\omega ct)e^{-i\omega x}\,d\omega$$

Recall that $\cos(x) = \frac{1}{2}\left(e^{ix} + e^{-ix}\right)$,

$$\begin{aligned} u(x,t) &= \frac{1}{2}\int_{-\infty}^{\infty} U(\omega,0)\left(e^{i\omega ct}+e^{-i\omega ct}\right)e^{-i\omega x}\,d\omega \\ &= \frac{1}{2}\int_{-\infty}^{\infty} U(\omega,0)\left[e^{-i\omega(x-ct)}+e^{-i\omega(x+ct)}\right]\,d\omega \end{aligned} \tag{5.2}$$

From the initial condition,

$$f(x) = u(x,0) = \mathscr{F}^{-1}\left[U(\omega,0)\right] = \int_{-\infty}^{\infty} U(\omega,0)e^{-i\omega x}\,d\omega$$

Eq. 5.2 is finally rewritten as,

$$\boxed{u(x,t) = \frac{1}{2}\left[f(x-ct)+f(x+ct)\right]}$$

Diffusion Equation with Convection

Consider the following problem,

$$\begin{aligned} \frac{\partial u}{\partial t} &= k\frac{\partial^2 u}{\partial x^2} + c\frac{\partial u}{\partial x}, \quad -\infty < x < \infty \\ u(x,0) &= f(x) \end{aligned}$$

Let us Fourier transform in x,

$$U(\omega,t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} u(x,t)e^{i\omega x}\,dx$$

The PDE becomes,

$$\begin{aligned} \frac{\partial U}{\partial t} &= k(-i\omega)^2 U + c(-i\omega)U \\ &= -\left[k\omega^2 + ic\omega\right]U \end{aligned}$$

The initial condition is transformed as well,

$$U(\omega,0) = F(\omega) = \frac{1}{2\pi}\int\limits_{-\infty}^{\infty} f(x)e^{i\omega x}\,dx$$

The solution to the ODE is,

$$\begin{aligned} U(\omega,t) &= F(\omega)e^{-(k\omega^2+ic\omega)t} \\ &= F(\omega)e^{-ic\omega t}e^{-k\omega^2 t} \end{aligned}$$

Recall that,

$$\mathscr{F}[g(x)] = e^{-k\omega^2 t}, \quad \text{where } g(x) = \sqrt{\frac{\pi}{kt}}\exp\left(-\frac{x^2}{4kt}\right)$$

The solution to the ODE is rewritten as,

$$\begin{aligned} U(\omega,t) &= \mathscr{F}\left[f(x)\right]e^{-ic\omega t}\mathscr{F}\left[g(x)\right] \\ &= \mathscr{F}[f(x)]\mathscr{F}[g(x+ct)] \end{aligned}$$

From the convolution theorem,

$$u(x,t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} f(\overline{x})g(x-\overline{x}+ct)\,d\overline{x}$$

$$\boxed{u(x,t) = \frac{1}{\sqrt{4\pi kt}}\int_{-\infty}^{\infty} f(\overline{x})\exp\left[-\frac{(x-\overline{x}+ct)^2}{4kt}\right]\,d\overline{x}}$$

5.4 Motivation: Sine and Cosine Transform

Let us solve the following problem using separation of variables,

$$\begin{aligned} \frac{\partial u}{\partial t} &= k\frac{\partial^2 u}{\partial x^2}, \quad x > 0 \\ \text{BC: } & u(0,t) = 0 \\ \text{IC: } & u(x,0) = f(x) \end{aligned}$$

Let's begin the separation of variables process,

$$u(x,t) = \phi(x)T(t)$$

Plug in the PDE,

$$\phi(x)T'(t) = k\phi''(x)T(t)$$

Divide by $k\phi T$,

$$\frac{T'(t)}{kT(t)} = \frac{\phi''(x)}{\phi(x)} = -\lambda$$

$$\begin{cases} T'(t) + \lambda kT = 0 \\ \phi''(x) + \lambda\phi(x) = 0, \quad \phi(0) = 0, \quad \lim_{x\to\infty} |\phi(x)| < \infty \end{cases}$$

For $\lambda > 0$,

$$\begin{aligned} \phi &= A\sin\left(x\sqrt{\lambda}\right) = A\sin(\omega x) \\ u &= Ae^{-k\omega^2 t}\sin(\omega x) \end{aligned}$$

Using the generalized principle of superposition,

$$\boxed{u(x,t) = \int_0^\infty A(\omega)e^{-k\omega^2 t}\sin(\omega x)\,d\omega}$$

From initial condition,

$$\begin{aligned} u(x,0) &= \int_0^\infty A(\omega)\sin(\omega x)\,d\omega = f(x) \\ A(\omega) &= \frac{2}{\pi}\int_0^\infty f(x)\sin(\omega x)\,dx \end{aligned}$$

5.5 Sine Transform

$$F_S(\omega) = \mathscr{F}_S[f(x)] = \frac{2}{\pi}\int_0^\infty f(x)\sin(\omega x)\,dx \quad \Longleftrightarrow \quad f(x) = \mathscr{F}_S^{-1}[F_S(\omega)] = \int_0^\infty F_S(\omega)\sin(\omega x)\,d\omega$$

$f(x)$	$F_S(\omega)$
$\dfrac{df}{dx}$	$-\omega \mathscr{F}_C\left[f(x)\right]$
$\dfrac{d^2f}{dx^2}$	$\dfrac{2}{\pi}\omega f(0) - \omega^2 \mathscr{F}_S[f(x)]$
1	$\dfrac{2}{\pi}\dfrac{1}{\omega}$
$\dfrac{1}{\pi}\int\limits_0^\infty f(\overline{x})\left[g(x-\overline{x}) - g(x+\overline{x})\right]\, d\overline{x}$	$\mathscr{F}_S[f(x)]\mathscr{F}_C[g(x)]$

5.6 Cosine Transform

$$F_C(\omega) = \mathscr{F}_C[f(x)] = \frac{2}{\pi}\int\limits_0^\infty f(x)\cos(\omega x)\, dx \quad \Longleftrightarrow \quad f(x) = \mathscr{F}_C^{-1}[F_C(\omega)] = \int\limits_0^\infty F_C(\omega)\cos(\omega x)\, d\omega$$

$f(x)$	$F_C(\omega)$
$\dfrac{df}{dx}$	$-\dfrac{2}{\pi}f(0) + \omega \mathscr{F}_S\left[f(x)\right]$
$\dfrac{d^2f}{dx^2}$	$-\dfrac{2}{\pi}f'(0) - \omega^2 \mathscr{F}_C\left[f(x)\right]$
$\exp\left[-\alpha x^2\right]$	$\dfrac{2}{\sqrt{4\pi\alpha}}\exp\left[-\dfrac{\omega^2}{4\alpha}\right]$

5.7 Sine/Cosine Transform Examples

Heat Equation on Semi-Infinite Interval

Consider the motivation problem,

$$\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}, \quad x > 0$$
$$\text{BC: } u(0,t) = 0$$
$$\text{IC: } u(x,0) = f(x)$$

We perform a Fourier sine transform in x,

$$U(\omega,t) = \frac{2}{\pi}\int_0^\infty u(x,t)\sin(\omega x)\,dx$$
$$\frac{\partial U}{\partial t} = k\left[\frac{2}{\pi}\omega u(0,t) - \omega^2 U\right]$$
$$\frac{\partial U}{\partial t} = -k\omega^2 U$$

Solve the differential equation and obtain,

$$U(\omega,t) = F(\omega)e^{-k\omega^2 t} = \mathscr{F}_s[f(x)]\mathscr{F}_c[g(x)], \quad \text{where } g(x) = \sqrt{\frac{\pi}{4kt}}\exp\left(-\frac{x^2}{4kt}\right)$$

Using the convolution theorem,

$$\boxed{u(x,t) = \frac{1}{\sqrt{4\pi kt}}\int_0^\infty f(\overline{x})\left\{\exp\left[-\frac{(x-\overline{x})^2}{4kt}\right] - \exp\left[-\frac{(x+\overline{x})^2}{4kt}\right]\right\}d\overline{x}}$$

Chapter 6

Laplace Transform

6.1 Laplace Transform Properties

Here are the definitions of the Laplace Transform,

$$F(s) = \mathscr{L}[f(t)] = \int_0^\infty f(t)\, e^{-st}\, dt \quad \Longleftrightarrow \quad f(t) = \mathscr{L}^{-1}[F] = \frac{1}{2\pi i} \int\limits_{c-i\infty}^{c+i\infty} F(s) e^{st}\, ds$$

$f(t)$	$F(s)$
$\dfrac{df}{dt}$	$sF(s) - f(0)$
$\dfrac{d^2 f}{dt^2}$	$s^2 F(s) - s f(0) - f'(0)$
$(f * g)(t) = \int_0^t f(\tau)\, g(t-\tau)\, d\tau$	$F(s)\, G(s)$
$\sinh(\alpha t)$	$\dfrac{\alpha}{s^2 - \alpha^2}$
$\cosh(\alpha t)$	$\dfrac{s}{s^2 - \alpha^2}$
$\delta(t-b)$	e^{-bs}

Chapter 7

Green's Functions

7.1 Motivation

Let's consider the following problem,

$$\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2} + Q(x,t)$$
$$\text{BCs: } u(0,t) = 0$$
$$u(L,t) = 0$$
$$\text{IC: } u(x,0) = g(x)$$

I solve this by the method of eigenfunction expansion.

$$Q(x,t) = \sum_{n=1}^{\infty} q_n(t)\sin\left(\frac{\pi n x}{L}\right)$$

Using orthogonality of sines, I can solve for the coefficients q_n,

$$q_n(t) = \frac{2}{L}\int_0^L Q(x,t)\sin\left(\frac{\pi n x}{L}\right)\,dx \tag{7.1}$$

The solution is in the form,

$$u(x,t) = \sum_{n=1}^{\infty} a_n(t)\sin\left(\frac{\pi n x}{L}\right) \tag{7.2}$$

Plug Eqs. 7.1 and 7.2 into the main governing PDE,

$$\sum_{n=1}^{\infty} a_n'(t)\sin\left(\frac{\pi n x}{L}\right) = k\sum_{n=1}^{\infty} -\left(\frac{\pi n}{L}\right)^2 a_n(t)\sin\left(\frac{\pi n x}{L}\right) + \sum_{n=1}^{\infty} q_n(t)\sin\left(\frac{\pi n x}{L}\right)$$
$$0 = \sum_{n=1}^{\infty}\left[a_n'(t) + k\left(\frac{\pi n}{L}\right)^2 a_n(t) - q_n(t)\right]\sin\left(\frac{\pi n x}{L}\right)$$

We have a differential equation for $a_n(t)$.

$$\frac{da_n}{dt} + k\left(\frac{\pi n}{L}\right)^2 a_n = q_n(t)$$

We solve this by multiplying by an integrating factor, $\exp[k(\pi n/L)^2 t]$,

$$\frac{d}{dt}\left[e^{k\left(\frac{\pi n}{L}\right)^2 t} a_n\right] = q_n(t)e^{k\left(\frac{\pi n}{L}\right)^2 t}$$

Take the anti-derivative of both sides,

$$a_n e^{k\left(\frac{\pi n}{L}\right)^2 t} = \int q_n(t)e^{k\left(\frac{\pi n}{L}\right)^2 t}\,dt$$
$$= \int_0^t q_n(t_0)e^{k\left(\frac{\pi n}{L}\right)^2 t_0}\,dt_0 + C$$
$$a_n(t) = e^{-k\left(\frac{\pi n}{L}\right)^2 t}\int_0^t q_n(t_0)e^{k\left(\frac{\pi n}{L}\right)^2 t_0}\,dt_0 + ce^{-k\left(\frac{\pi n}{L}\right)^2 t}$$

The constant is $c = a_n(0)$.

Eq. 7.2 becomes, with $a_n(t)$ plugged in,

$$\boxed{u(x,t) = \sum_{n=1}^{\infty}\left\{\left[a_n(0)e^{-k\left(\frac{\pi n}{L}\right)^2 t} + e^{-k\left(\frac{\pi n}{L}\right)^2 t}\int_0^t q_n(t_0)e^{k\left(\frac{\pi n}{L}\right)^2 t_0}\,dt_0\right]\sin\left(\frac{\pi n x}{L}\right)\right\}}$$

From the initial condition,

$$u(x,0) = \sum_{n=1}^{\infty} a_n(0)\sin\left(\frac{\pi n x}{L}\right) = g(x)$$

Solve for $a_n(0)$ by using orthogonality relations,

$$\int_0^L \sum_{n=1}^{\infty} a_n(0)\sin\left(\frac{\pi n x}{L}\right)\sin\left(\frac{\pi m x}{L}\right)\,dx = \int_0^L g(x)\sin\left(\frac{\pi m x}{L}\right)\,dx$$
$$a_n(0) = \frac{2}{L}\int_0^L g(x)\sin\left(\frac{\pi n x}{L}\right)\,dx \tag{7.3}$$

I plug Eqs. 7.1 and Eq. 7.3 into the boxed equation,

$$
\begin{aligned}
u(x,t) = \sum_{n=1}^{\infty} \Bigg\{ & \left[\frac{2}{L} \int_0^L g(x_0) \sin\left(\frac{\pi n x_0}{L}\right) \, dx_0 \right] e^{-k\left(\frac{\pi n}{L}\right)^2 t} \\
& + e^{-k\left(\frac{\pi n}{L}\right)^2 t} \int_0^t \left[\frac{2}{L} \int_0^L Q(x_0, t_0) \sin\left(\frac{\pi n x_0}{L}\right) \, dx_0 \right] e^{k\left(\frac{\pi n}{L}\right)^2 t_0} \, dt_0 \Bigg\} \sin\left(\frac{\pi n x}{L}\right)
\end{aligned}
$$

$$
\boxed{
\begin{aligned}
u(x,t) = & \int_0^L g(x_0) \left[\sum_{n=1}^{\infty} \frac{2}{L} \sin\left(\frac{n\pi x_0}{L}\right) \sin\left(\frac{\pi n x}{L}\right) e^{-k\left(\frac{\pi n}{L}\right)^2 t} \right] dx_0 \\
& + \int_0^L \int_0^t Q(x_0, t_0) \left[\sum_{n=1}^{\infty} \frac{2}{L} \sin\left(\frac{n\pi x_0}{L}\right) \sin\left(\frac{n\pi x}{L}\right) e^{-k\left(\frac{\pi n}{L}\right)^2 (t - t_0)} \right] dt_0 dx_0
\end{aligned}
}
$$

The term in brackets is the **Green's function**. Using the notation,

$$
\boxed{G(x, t; x_0, t_0) = \sum_{n=1}^{\infty} \frac{2}{L} \sin\left(\frac{n\pi x_0}{L}\right) \sin\left(\frac{n\pi x}{L}\right) e^{-k\left(\frac{n\pi}{L}\right)^2 (t - t_0)}}
$$

We rewrite the solution as,

$$
\boxed{u(x,t) = \int_0^L g(x_0) G(x, t; x_0, 0) \, dx_0 + \int_0^L \int_0^t Q(x_0, t_0) G(x, t; x_0, t_0) \, dt_0 dx_0}
$$

7.2 Three Methods of Getting Green's Function

7.2.1 Variation of Parameters Method

Variation of Parameters Method: Green's Function for Simple ODE

Consider the following ODE,

$$
\begin{aligned}
\frac{d^2 u}{dx^2} &= f(x) \\
u(0) &= 0 \\
u(L) &= 0
\end{aligned}
$$

According to variation of parameters, when we have two homogeneous solutions u_1 and u_2, a particular solution is,

$$u(x) = v_1(x)u_1(x) + v_2(x)u_2(x)$$

where,

$$\begin{aligned}
\frac{dv_1}{dx} &= -\frac{f(x)u_2}{c} \\
\frac{dv_2}{dx} &= \frac{f(x)u_1}{c} \\
c &= u_1\frac{du_2}{dx} - u_2\frac{du_1}{dx}
\end{aligned}$$

We choose the two homogeneous solutions to be,

$$\begin{aligned}
u_1 &= x \\
u_2 &= L - x
\end{aligned}$$

, where u_1 satisfies the first boundary condition and u_2 satisfies the second boundary condition.

In our particular case,

$$\begin{aligned}
c &= -L \\
\frac{dv_1}{dx} &= \frac{(L-x)f(x)}{L} \\
\frac{dv_2}{dx} &= -\frac{xf(x)}{L}
\end{aligned}$$

Integration,

$$\begin{aligned}
v_1(x) &= \frac{1}{L}\int_0^x (L-x_0)f(x_0)\,dx_0 + c_1 \\
v_2(x) &= -\frac{1}{L}\int_0^x x_0 f(x_0)\,dx_0 + c_2
\end{aligned}$$

We can solve for the two constants c_1 and c_2 by using the boundary conditions,

$$\begin{aligned}
u(0) = 0 &\quad\longrightarrow\quad c_2 = 0 \\
u(L) = 0 &\quad\longrightarrow\quad c_1 = -\frac{1}{L}\int_0^L (L-x_0)f(x_0)\,dx_0
\end{aligned}$$

Plugging in these constants into $v_1(x)$ and $v_2(x)$,

$$v_1(x) = \frac{1}{L}\int_L^x (L - x_0) f(x_0)\, dx_0$$
$$v_2(x) = -\frac{1}{L}\int_0^x x_0 f(x_0)\, dx_0$$

The solution is therefore,

$$u = -\frac{x}{L}\int_x^L (L - x_0) f(x_0)\, dx_0 - \frac{(L-x)}{L}\int_0^x x_0 f(x_0)\, dx_0$$

Rewrite in the form,

$$\boxed{\begin{aligned} u(x) &= \int_0^L f(x_0) G(x, x_0)\, dx_0 \\ G(x, x_0) &= \begin{cases} \dfrac{-x(L - x_0)}{L}, & x < x_0 \\ \dfrac{-x_0(L - x)}{L}, & x > x_0 \end{cases} \end{aligned}}$$

7.2.2 Eigenfunction Expansion Method: Green's Function

Suppose we have,

$$L(u) = f(x)$$

where L is the Sturm-Liouville operator.

The associated eigenvalue problem is,

$$L(\phi) = -\lambda\sigma\phi$$

We know from eigenfunction expansion theory that the solution can be written as,

$$u(x) = \sum_{n=1}^{\infty} a_n \phi_n(x)$$

If we plug back into the original problem,

$$\sum_{n=1}^{\infty} a_n L(\phi_n) = \sum_{n=1}^{\infty} -a_n \lambda_n \sigma \phi_n = f(x)$$

Using orthogonality of the eigenfunctions,

$$-a_n \lambda_n \int_a^b \sigma \phi_n^2 \, dx = \int_a^b f(x) \phi_n \, dx$$
$$a_n = \frac{-\int_a^b f(x)\phi_n(x)\, dx}{\lambda_n \int_a^b \sigma \phi_n(x)^2 \, dx}$$

The solution is thus,

$$\boxed{u = \int_a^b f(x_0) \sum_{n=1}^{\infty} \frac{\phi_n(x)\phi_n(x_0)}{-\lambda_n \int_a^b \phi_n^2 \sigma \, dx} \, dx_0 = \int_a^b f(x_0) G(x, x_0)\, dx_0}$$

where

$$\boxed{G(x, x_0) = \sum_{n=1}^{\infty} \frac{\phi_n(x)\phi_n(x_0)}{-\lambda_n \int_a^b \phi_n^2 \sigma \, dx}}$$

Eigenfunction Expansion Method: Green's Function for Simple ODE

Consider the following ODE,

$$\frac{d^2 u}{dx^2} = f(x)$$
$$u(0) = 0$$
$$u(L) = 0$$

The related eigenvalue problem is,

$$\frac{d^2 \phi}{dx^2} = -\lambda \phi$$
$$\phi(0) = 0$$
$$\phi(L) = 0$$

The solution is,

$$\phi_n = \sin\left(\frac{\pi n x}{L}\right)$$

The solution to the original ODE is in the form,

$$u = \sum_n a_n \phi_n(x)$$

Plug into ODE,

$$\sum_{n=1}^{\infty} -a_n \left(\frac{\pi n}{L}\right)^2 \sin\left(\frac{\pi n x}{L}\right) = f(x)$$
$$\int_0^L \sum_{n=1}^{\infty} -a_n \left(\frac{\pi n}{L}\right)^2 \sin\left(\frac{\pi n x}{L}\right) \sin\left(\frac{\pi m x}{L}\right)\, dx = \int_0^L f(x) \sin\left(\frac{\pi m x}{L}\right)\, dx$$
$$a_n = -\frac{2}{L} \frac{\int_0^L f(x_0) \sin\left(\frac{\pi n x_0}{L}\right)\, dx_0}{(\pi n/L)^2}$$

Therefore, the solution can be written as,

$$\boxed{\begin{aligned} u &= \int_0^L f(x_0) \sum_n \frac{-\frac{2}{L}\sin\left(\frac{\pi n x}{L}\right)\sin\left(\frac{\pi n x_0}{L}\right)}{(\pi n/L)^2}\, dx_0 \\ u &= \int_0^L f(x_0) G(x, x_0)\, dx_0 \end{aligned}}$$

7.2.3 Dirac-Delta Method

Dirac-Delta Method: Green's Function for Simple ODE

Consider the following ODE,

$$\frac{d^2u}{dx^2} = f(x)$$
$$u(0) = 0$$
$$u(L) = 0$$

We first calculate Green's function by solving the following ODE,

$$\begin{aligned}\frac{d^2G(x,x_0)}{dx^2} &= \delta(x-x_0)\\ G(0,x_0) &= 0\\ G(L,x_0) &= 0\end{aligned}$$

For x before and after x_0, this is equivalent to solving for the homogeneous case, $d^2G/dx^2 = 0$.

$$G(x,x_0) = \begin{cases} a+bx, & x < x_0 \\ c+dx, & x > x_0 \end{cases}$$

Applying the boundary condition, we get that $a = 0, \quad c = -dL$,

$$G(x,x_0) = \begin{cases} bx, & x < x_0 \\ d(x-L), & x > x_0 \end{cases}$$

To solve for b and d, we use two additional constraints: one arising from continuity and the other arising from integrating the Green's function ODE,

$$\left\{\begin{aligned} G(x_0^-,x_0) &= G(x_0^+,x_0) \\ \left.\frac{dG}{dx}\right|_{x=x_0^+} - \left.\frac{dG}{dx}\right|_{x=x_0^-} &= 1 \end{aligned}\right\}$$

$$\left\{\begin{aligned} bx_0 &= d(x_0-L) \\ d-b &= 1 \end{aligned}\right\}$$

Solving, we get $d = x_0/L$ and $b = (x_0-L)/L$

$$\boxed{G(x,x_0) = \left\{\begin{aligned} -\frac{x}{L}\left(L-x_0\right), &\quad x < x_0 \\ -\frac{x_0}{L}(L-x), &\quad x > x_0 \end{aligned}\right\}}$$